

JAEHWAN LEE

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RESEARCH INTERESTS

AI Security, Fairness, AI Applications on Industry Systems, Generative AI, Domain Generalization, Anomaly Detection

EDUCATION

Dongguk University Seoul, Korea
Master of Engineering in Industrial and Systems Engineering Sep 2021-Aug 2023
Cumulative GPA: 4.33/4.5

Dongguk University Seoul, Korea
Bachelor of Science in Industrial and Systems Engineering Mar 2016-Aug 2021
Cumulative GPA: 4.03/4.5

RESEARCH EXPERIENCE

AhnLab, Inc. Seongnam, Korea
AhnLab, Inc. is a prominent cybersecurity firm located in South Korea that focuses on delivering a wide range of security solutions, including antivirus software, network security, and endpoint protection. Currently, Ahnlab, Inc. is focusing on developing risk detection systems and security solutions by integrating AI and machine learning.

AI Researcher, Division of Core Development Jun 2023-Jul 2024

- Developed malware detection models for IT security, enhancing threat identification and response capabilities
 - Created a Python library to identify a suitable representation for Microsoft files such as Word, Excel, and PowerPoint that can contain viruses or malicious macros
 - Utilized PySpark for feature engineering and developed a model to classify malicious files
 - Established an automated training pipeline using large-scale ML and orchestration tools like SynapseML, Kubeflow, MLflow and KServe, enabling monthly model updates
- Created an AI security assistant utilizing open-source large language models (LLMs) to improve automated security support and incident management.
 - Collected and processed the cybersecurity domain data, fine-tuned various open-source LLMs to enable tasks such as NLU, multiple choice answering, and function calling
 - Evaluated model performance using Evaluation-Harness, deploying the top-performing model on a TGI inference server for internal use.

Dongguk University Seoul, Korea
Research Assistant, Data Science Laboratory (Supervisor: Professor Youngdoo Son) Jun 2021-Jun 2023

Project: Domain Generalization Using Training and Test Time Augmentation in Frequency Spaces (Sep 2022-Jun 2023)

- Investigated applications of 2D Fourier transforms to enhance domain generalization tasks
 - Proposed a robust data augmentation technique that preserves the phase information of images in the frequency domain while blending amplitude information through linear interpolation of images from different domains
 - Implemented a memory bank to store the most invariant style amplitudes, enabling the application of test time augmentation (TTA) during the inference phase
- Discovered that style variations can be effectively derived by altering the amplitude information that expresses the style while preserving the essence represented in the frequency space of an image, thereby outperforming existing augmentation techniques

Project: Design of BNN-Based Computer Vision Sensor for On-Device Facial Recognition (Jun 2021-Apr 2022)

- Conducted research to create a low-power, always-on facial recognition sensor by training a convolutional neural network (CNN) model with binary parameters (0s and 1s)
- Involved in enhancing the model's focus on specific bits through non-linear quantization during the image processing phase

- Explored advancements in integrating neural network processing into CMOS image sensors, reducing the need for separate external processors, thereby lowering power consumption and improving data processing efficiency
- Recognized the value of quantization techniques, such as nonlinear quantization and weight binarization, which can maintain high accuracy while minimizing hardware footprint and energy consumption

Project: Development of Explainable Meta Learning (X-META) Methodology and Decision Support System for Facility-Agnostic Prognostics and Health Management Funded by the National Research Foundation of Korea (Mar 2021-Mar 2023)

- Developed a representation learning technique for generalized fault diagnosis of equipment, ensuring class consistency
 - Trained similar samples to cluster together while pushing dissimilar samples apart and incorporated a loss function that reflected intra-class similarity, allowing for the extraction of graph features that were highly effective for fault classification
- Proposed a graph-based feature extraction and fault diagnosis technique that incorporates global structural information from time series data
 - Constructed adjacency matrices using both Euclidean distance and visibility algorithms, which related to the connectivity strength and connectivity of the time series, followed by the computation of the Frobenius norm of each adjacency matrix
 - Verified that the proposed features exhibited well-separated distributions based on bearing conditions and confirmed that the features could be meaningfully utilized for fault diagnosis

Project: Development of AI-Based High-Efficiency Water Electrolysis Catalyst Funded by the National Research Foundation of Korea (Dec 2021-Dec 2022)

- Conducted research on efficient machine learning methods leveraging descriptor searches to mitigate the computational complexity and processing time associated with Density Functional Theory, aimed at predicting properties for evaluating the suitability of 2D materials as water electrolysis catalysts
 - Utilized genetic algorithms to derive a set of descriptors that can best represent each compound
 - Designed the process to preserve genetic diversity while iteratively reducing the residuals of the target chemical properties, facilitating the efficient and gradual identification of global optima
- Theoretically and visually demonstrated that the descriptors obtained through this proposed method, which consist of nonlinear combinations of primary features such as atomic mass and valence electrons, are capable of sufficiently representing the properties of complex compounds

Project: Benign and Malignant Prediction Using Gastrointestinal Pathology Images Funded by the Ministry of Science and Information and Communication Technology (ICT), Government of the Republic of Korea (Sep 2021-Feb 2022)

- Leveraged specific data augmentation techniques tailored for pathology images and implemented model quantization to ensure real-time applicability
- Gained valuable insights into the importance of employing specialized data augmentation techniques that account for the specific characteristics of the unique domain, such as pathology images
- Learned how to enhance both model performance and efficiency by applying various quantization techniques and optimizing the models through lightweighting

PUBLICATIONS

Lee, SH., Lee, S., **Lee, J.**, Lee, W., and Son, Y. Graph Contrastive Learning with Consistency Regularization. *Pattern Recognition Letters*. May 2024. doi: 10.1016/j.patrec.2024.03.014

Lee, JH., **Lee, J.**, Shin, S., Son, Y., and Han, YK. Machine Learning for the Expedited Screening of Hydrogen Evolution Catalysts for Transition Metal-Doped Transition Metal Dichalcogenides. *International Journal of Energy Research*. September 2023. doi: 10.1155/2023/6612054

Lee, J., Shin, S., Lee, J., Han, YK., Lee, W., and Son, Y. Genetic Descriptor Search Algorithm for Predicting Hydrogen Adsorption Free Energy of 2D Material. *Scientific Reports*. August 2023. doi: 10.1038/s41598-023-39696-0

Jeong, B., **Lee, J.**, Lee, S., Lee, SY., Son, Y., and Kim, S. A 240 FPS In-Column Binarized Neural Network Processing in CMOS Image Sensors. *IEEE Transactions on Circuits and Systems II: Express Briefs*. July 2023. doi: 10.1109/TCSII.2023.3295391

Jeong, B., **Lee, J.**, Choi, J., Song, M., Son, Y., and Kim, S. A 0.57 mW@ 1 FPS In-Column Analog CNN Processor Integrated into CMOS Image Sensor. *IEEE Access*. June 2023. doi: 10.1109/ACCESS.2023.3286544

PRESENTATIONS

Lee, J., and Son, Y. Domain Generalization Using Phase Consistency. *Presented at the Korea Institute of Industrial Engineers Fall Conference*. Incheon, Korea. November 2022.

Shin, S., **Lee, J.**, Choi, C., and Son, Y. Task Balancing in Multitask learning Using Meta-Learning. *Presented at the Korea Institute of Industrial Engineers Spring Joint Conference*. Jeju, Korea. June 2022.

Lee, J., Park, H., and Son, Y. A Genetic Algorithm for 2D Material Property Prediction. *Presented at the Korea Institute of Industrial Engineers Fall Conference*. Seoul, Korea. November 2021.

Lee, S., Lee, SH., Hong, J., **Lee, J.**, and Son, Y. Frequency Based Graph Feature Extraction and Fault Detection. *Presented at the Korea Institute of Industrial Engineers Spring Joint Conference*. Jeju, Korea. June 2021

TEACHING EXPERIENCE

Dongguk University

Seoul, Korea

Teaching Assistant, Department of Industrial & Systems Engineering

Sep 2020-Sep 2022

- Created and prepared instructional materials for classroom teaching
- Conducted coding exercises during lab sessions to reinforce theoretical concepts and provided support through question-and-answer sessions during and after lab hours
- Organized and supervised examinations and graded assignments

Courses: Advanced Manufacturing Processes (2nd Semester 2022), Adventure Design (2nd Semester 2022), Introduction to Machine Learning (1st Semester 2021, 1st Semester 2022), Introduction to Financial Engineering (1st Semester 2021, 1st Semester 2022), Advanced Data Science (1st Semester 2022), Introduction to Data Science (2nd Semester 2020)

HONORS AND AWARDS

Teaching Assistant Scholarship (\$4,700), Dongguk University

Mar 2022, Sep 2022

Honorable Award, National Pathology Health Datathon

Nov 2021

Dongguk Future Talent Scholarship (\$12,300), Dongguk University

Mar 2021, Sep 2021

Honors Graduation Award, Dongguk University

Aug 2021

Course Top Scholarship (\$289), Dongguk University

Sep 2019, Sep 2020, Mar 2021

- Probability and Statistics, Operation Research II, Simulation and Applications, Industrial AI

Academic Excellence Scholarship (\$2,890), Dongguk University

Sep 2019, Mar 2020, Mar 2021

Honorable Award, Dongguk Entrepreneur Project Competition

Feb 2017

MILITARY SERVICE

Korean Augmentation to the US Army, Yongsan Garrison and Camp Humphreys

Seoul, Korea

Discharged as a Sergeant

Mar 2018-Dec 2019

- Supervised and maintained all Army supplies and equipment for the Joint United States Military Affairs Group-Korea (JUSMAG-K)
- Served as Senior KATUSA, promoting cooperation between U.S. and Korean forces, and leading KATUSA soldiers.
- Key responsibilities included providing military supplies during Ulchi Freedom Guardian (UFG) and Key Resolve (KR) exercises

TECHNICAL SKILLS

Computer Languages: Python, R, Markdown, LaTeX

Frameworks: PyTorch, PySpark, vLLM, Kubernetes, Linux, MLflow, Kuberflow, MinIO, Github